

ALEXANDER HEWITT

Data Scientist | Predictive Maintenance & Logistics Analytics

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CLEARANCE & CITIZENSHIP

U.S. Citizen. Tier 5 (Top Secret/SCI-equivalent) background investigation favorably adjudicated in 2025. No clearance is currently active because no need-to-know exists in my present position; I am eligible for Secret sponsorship/reinstatement and previously served on-site in a cleared Department of the Air Force depot environment (Tinker AFB, 2011–2014 and 2020–2022).

PROFESSIONAL SUMMARY

Data scientist and operations research analyst with 10+ years applying predictive modeling, statistical analysis, and ETL/data engineering to U.S. Air Force aircraft sustainment and federal forecasting. As lead analyst for the 565th Aircraft Maintenance Squadron (B-52, Tinker AFB) I designed a Functional Check Flight defect tracking program covering 700+ maintenance events, identified recurring failure patterns across 10+ parts, and cut post-dock flow days by 5+ days (~10%) — the same condition-based maintenance and reliability problem set this role supports. Currently lead \$7B+ revenue and cost forecasting models at USCIS in Databricks, PySpark, SQL, and Python, with a Master of Science in Data Science (4.0 GPA) and Lean Six Sigma Green Belt.

TECHNICAL SKILLS

- **Analytics & Data Engineering:** Databricks, PySpark, SQL (advanced), Python (pandas, NumPy, scikit-learn, XGBoost, LightGBM), R, SAS; ETL pipeline implementation, sustainment, and optimization; data standardization and governance; MLflow; technical documentation standards.
- **Statistics & Modeling:** Statistical distributions and applied probability, hypothesis testing, confidence intervals, Monte Carlo simulation, time-series forecasting, scenario modeling, feature engineering, predictive modeling, model performance monitoring and drift detection.
- **NLP & Text Analytics:** Applied natural language processing coursework (tokenization, text classification, entity/keyword extraction, embeddings/transformer models); data labeling and annotation workflow design for structured extraction.
- **Visualization & Productivity:** Qlik Sense, Streamlit, Tableau, Power BI, Excel (VBA/macros), Jira, Confluence, Lucidchart, Microsoft Office Suite (Word, Excel, PowerPoint).
- **Sustainment Domain:** CBM+/predictive maintenance analytics, depot production flow, aircraft maintenance and equipment health indicators, manpower and labor-capacity analysis, bottleneck and root cause analysis, Lean Six Sigma, program performance analysis, defense logistics.

PROFESSIONAL EXPERIENCE

Operations Research Analyst (GS-14)

August 2022 – Present

U.S. Citizenship and Immigration Services (USCIS) — Remote

- Lead revenue and cost forecasting across 30+ revenue-generating forms supporting analysis of more than \$7B annually using Python, SQL, SAS, Databricks, and Tableau.
- Oversee end-to-end data collection and processing: build, sustain, and optimize ETL pipelines that standardize and integrate large, interrelated datasets for modeling, outlier detection, and leadership decision support; automated recurring reporting processes saving ~24 workdays per year.
- Perform exploratory data analysis, statistical analysis, and data visualization to surface trends, correlations, and actionable insights that drive program-wide product development.
- Continuously monitor data quality and model performance to detect and mitigate drift and degradation; document data governance standards and methodologies.
- Translate complex technical findings into operational and financial recommendations for non-technical and executive audiences, and collaborate with cross-functional engineering teams to integrate models into production reporting and applications.

B-52 Aircraft Lead Data / Management Analyst (GS-12)

August 2021 – August 2022

U.S. Air Force, Tinker Air Force Base — Oklahoma City, OK

- Lead analyst for aircraft maintenance operations, program effectiveness, manpower analysis, and production decision-making for the 565th Aircraft Maintenance Squadron.
- Designed and implemented a Functional Check Flight (FCF) defect tracking program covering 700+ events; identified recurring failure patterns in 10+ parts, establishing the reliability and equipment-health signal that reduced post-dock flow days by 5+ days (~10%).
- Conducted statistical and operational analysis on maintenance, parts, and manpower data to identify bottlenecks and improve production flow — hands-on sustainment analytics directly transferable to CBM+ condition-based maintenance initiatives.
- Quantified the projected impact of policy changes on labor capacity, identifying risk of 35,000 lost production hours (~7% of annual target output), and briefed mitigation options to senior leadership.
- Produced dashboards, reports, and visual summaries for senior leaders on readiness, risk mitigation, and resource allocation; partnered across maintenance, planning, scheduling, and material teams to align analysis with operational constraints.

B-52 Aircraft Maintenance Data Analyst (GS-11)**August 2020 – August 2021***U.S. Air Force, Tinker Air Force Base — Oklahoma City, OK*

- Evaluated maintenance operations, production metrics, and workforce performance to support mission-critical aircraft sustainment.
- Performed root cause and bottleneck analysis across multiple functional areas including avionics, sheet metal, and electrics; built and improved recurring reports and Excel macro-based processes to reduce manual effort and improve reporting accuracy.
- Supported squadron leadership with targeted analysis of current- and future-state operations, constraints, and performance trends.

Data Analyst / Maintenance Scheduler (GS-11)**July 2011 – January 2014***U.S. Air Force, Tinker Air Force Base — Oklahoma City, OK*

- Created performance and efficiency reports for maintenance teams and skill groups supporting aircraft operations, establishing a baseline for labor-hour accounting and shop-floor performance measurement.
- Initiated a labeling system for 3,000+ part numbers, improving part search efficiency and shop-floor data accuracy.
- Led and supported 10+ process re-engineering efforts tied to maintenance and operational performance.

Security Services Operations Specialist I**March 2020 – August 2020***Bank of Oklahoma — Tulsa, OK*

- Managed transactional processing and reconciliations involving more than \$30M in daily client funds with a high level of accuracy and control; supported audit, compliance, and operational stakeholders through detailed reporting and issue resolution.

Day Trader / Portfolio Analyst**April 2017 – March 2020***HFLLC — Tulsa, OK*

- Managed and analyzed high-volume derivatives activity totaling \$15M+ in a calendar year; developed rule-based, time-series decision models and maintained a Sharpe Ratio of 1.2 through disciplined, probabilistic risk-based analysis.

Real Estate Investor / Analyst**January 2014 – April 2017***Self-Employed — Tulsa, OK*

- Conducted cost-benefit analysis on 50+ properties and managed 15+ projects across acquisition, renovation, and resale; reduced contractor costs by 20% through data-informed vendor selection and project planning.

EDUCATION

Master of Science in Data Science — GPA 4.0**March 2026***Eastern University*

- Capstone: end-to-end regression pipeline over ~5.5M flight records joined to hourly weather data — data acquisition, cleaning, multi-key integration, feature engineering, six models trained with hyperparameter search (LightGBM, XGBoost, Random Forest, Ridge, Lasso, OLS), residual and feature-importance analysis, and deployment through an interactive Streamlit application.
- Relevant coursework: Fundamentals of Machine Learning, Statistical Modeling, Data Manipulation, Data & Database Management with SQL, Information Visualization, Applied Data Science, AI Ethics.

Bachelor of Business Administration, Supply Chain Management**May 2011***University of Oklahoma***CERTIFICATIONS & TRAINING**

- Lean Six Sigma Green Belt
- DataCamp coursework: natural language processing, machine learning, deep learning, Python, SQL, Tableau, R